9.C Graph Reciprocal Functions

Reciprocal functions are defined by division and behave differently from polynomial functions. Their graphs are shaped by asymptotes and symmetry rather than intercepts and turning points.

> *"Reciprocal graphs are defined by what they approach, not what they reach."*

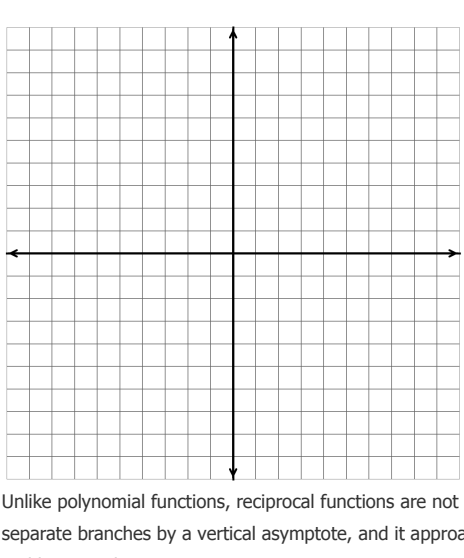
Learning Targets

I can graph basic reciprocal functions and identify key features.
I can identify asymptotes and intercepts.
I can describe how transformations affect reciprocal graphs.

Introduction

The parent reciprocal function is:

$$f(x) = \frac{1}{x}$$



Unlike polynomial functions, reciprocal functions are not continuous. The graph is split into separate branches by a vertical asymptote, and it approaches—but never touches—both vertical and horizontal asymptotes.

Key Idea: Reciprocal functions are defined by asymptotes and symmetry, not by intercepts or turning points.

Connections

Prior Knowledge:
- Transformations of functions from 9.A
- Asymptotes from 9.B
- Understanding domain and restrictions

Cross-Disciplinary Connections:
- Physics: light intensity decreases as distance increases (inverse relationship)
- Chemistry: concentration changes relative to volume in solutions
- Computer Science: time per task decreases as resources increase

Real-World Applications:
- Cost per person when splitting a bill: $\text{cost} = \frac{\text{total}}{x}$
- Time per worker when completing a job: $\text{time} = \frac{\text{work}}{x}$
- Travel time for a fixed distance: $\text{time} = \frac{d}{v}$
- Signal strength or brightness decreasing with distance
- Density relationships: $\text{density} = \frac{\text{mass}}{\text{volume}}$

History

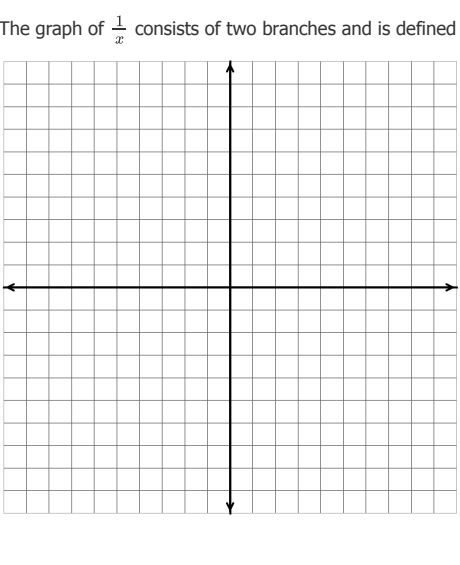
Reciprocal relationships have been studied since ancient mathematics, where ratios were used to compare quantities. These early ideas laid the foundation for functions defined by division.
With the development of coordinate geometry in the 17th century, mathematicians such as René Descartes began graphing functions like $\frac{1}{x}$, revealing their distinct structure and asymptotic behavior.
Today, reciprocal functions are used in science and engineering to model inverse relationships and systems that approach limits without reaching them.

Key Concepts

Parent Reciprocal Function

$$f(x) = \frac{1}{x}$$

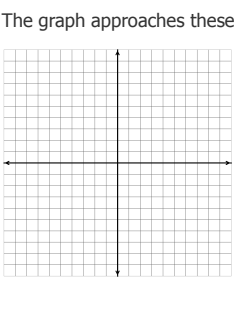
The graph of $\frac{1}{x}$ consists of two branches and is defined for all real values except $x = 0$.



Asymptotes

Reciprocal functions have two asymptotes that define their structure:
- **Vertical Asymptote:** $x = 0$
- **Horizontal Asymptote:** $y = 0$

The graph approaches these lines but never touches them.



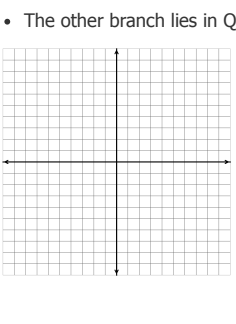
Intercepts

Reciprocal functions do not have an x-intercept because the function never equals zero.
There is also no y-intercept because the function is undefined at $x = 0$.

Key Observation:
The graph of a reciprocal function has no intercepts.

Symmetry and Orientation

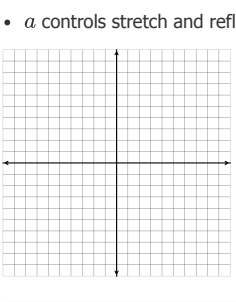
The graph of $\frac{1}{x}$ is symmetric about the origin.
- One branch lies in Quadrant I
- The other branch lies in Quadrant III



Transformations

$$f(x) = \frac{a}{x-h} + k$$

- h shifts the graph left or right
- k shifts the graph up or down
- a controls stretch and reflection



Key Idea: Reciprocal graphs are defined by asymptotes and symmetry. Transformations move the graph but do not change its fundamental structure.

Vocabulary

Reciprocal Function:
A function of the form $f(x) = \frac{1}{x}$, where the variable appears in the denominator.

Asymptote:
A line that the graph approaches but never touches.

Vertical Asymptote:
A vertical line where the function is undefined and the graph approaches without crossing.

Horizontal Asymptote:
A horizontal line that describes the end behavior of the function as x becomes large.

Symmetry:
A property where a graph is unchanged under a transformation, such as reflection about the origin.

Branch:
One part of a graph separated by a vertical asymptote.

Strategies

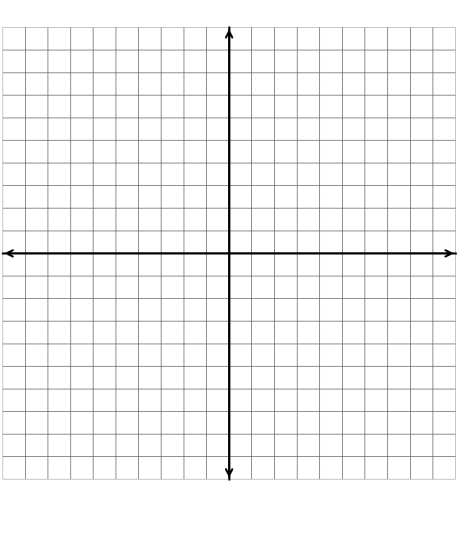
Graphing Reciprocal Functions:
1. **Identify asymptotes first.**
   Determine the vertical and horizontal asymptotes before drawing the graph.
2. **Determine orientation.**
   Use the sign of the function to decide which quadrants the branches will lie in.
3. **Apply transformations.**
   Shift the graph according to h and k, and adjust for reflections or stretches using a.
4. **Sketch branches near asymptotes.**
   Draw the graph approaching the asymptotes without touching them.
5. **Check for intercepts.**
   Verify whether intercepts exist based on the structure of the function.

Examples

Example 1 — Parent Function

Identify asymptotes and describe the graph.

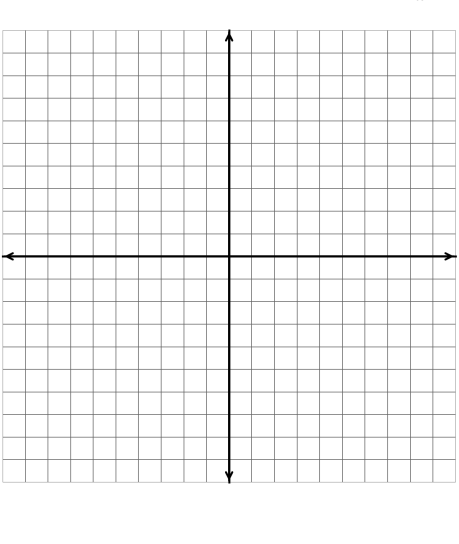
$$f(x) = \frac{1}{x}$$



Example 2 — Horizontal Shift

Identify asymptotes and sketch the graph.

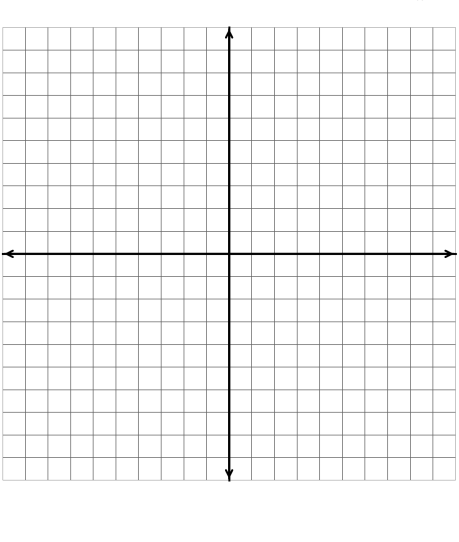
$$f(x) = \frac{1}{x-2}$$



Example 3 — Vertical Shift

Identify asymptotes and sketch the graph.

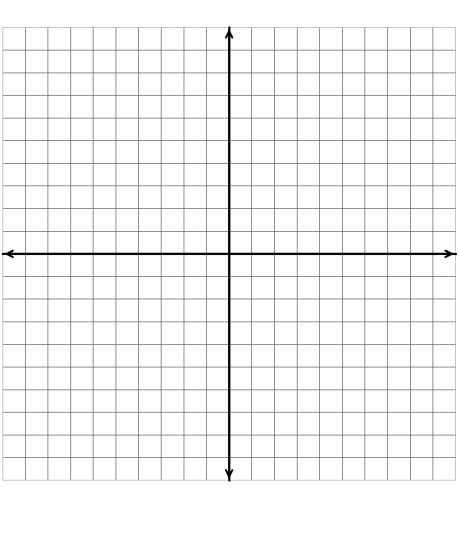
$$f(x) = \frac{1}{x} + 3$$



Example 4 — Reflection

Determine the orientation of the graph.

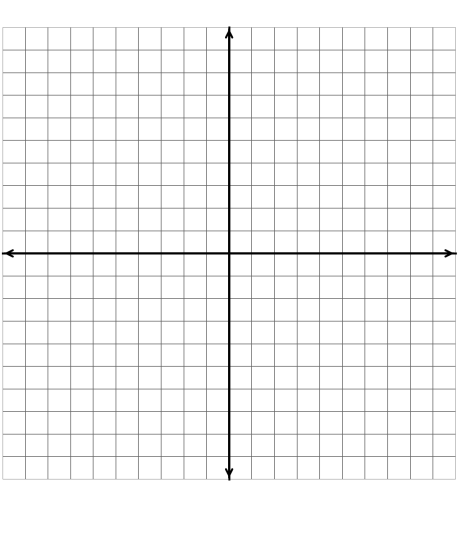
$$f(x) = -\frac{1}{x}$$



Example 5 — Vertical Stretch

Describe how the coefficient affects the graph.

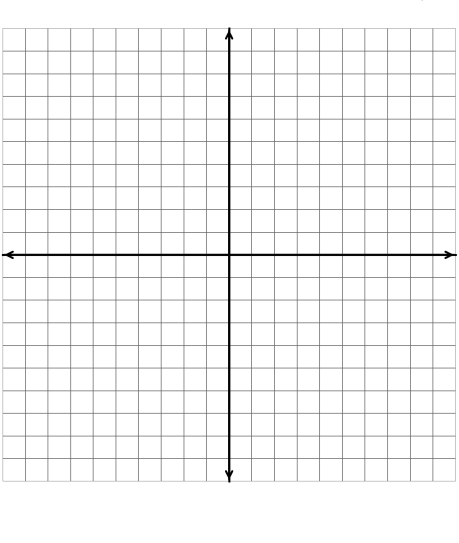
$$f(x) = \frac{2}{x}$$



Example 6 — Combined Transformations

Identify asymptotes and describe all transformations.

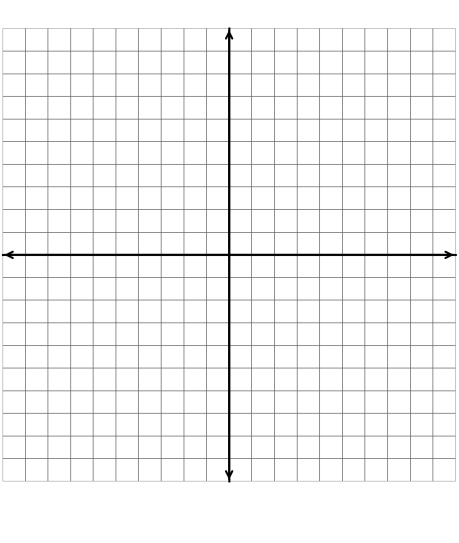
$$f(x) = \frac{1}{x+4} - 1$$



Example 7 — Reflection and Translation

Determine asymptotes, orientation, and transformations.

$$f(x) = -\frac{3}{x-1} + 2$$



Common Mistakes

- **Crossing asymptotes:**
  Reciprocal graphs approach asymptotes but never touch or cross them.
- **Incorrect orientation:**
  Forgetting that a negative sign reflects the graph across the x-axis, changing which quadrants the branches occupy.
- **Misidentifying asymptotes:**
  Students often forget to shift asymptotes when h and k are present.
- **Expecting intercepts:**
  Reciprocal functions typically do not have x- or y-intercepts.
- **Graphing like a polynomial:**
  Reciprocal functions are not continuous curves—they have separate branches.

Summary

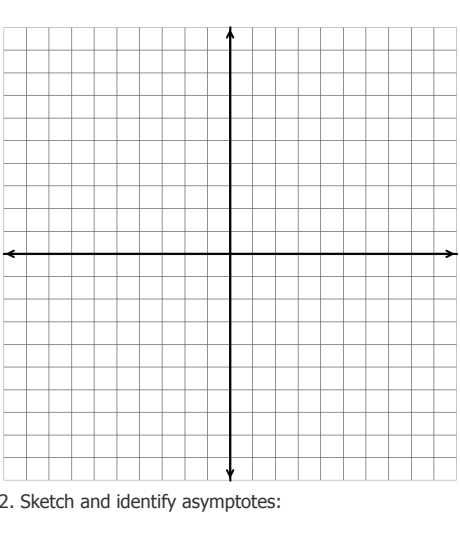
- Reciprocal functions are defined by asymptotes and symmetry.
- The parent function $frac1x$ has asymptotes at $x = 0$ and $y = 0$.
- Transformations shift asymptotes and change orientation.
- The graph consists of two branches separated by a vertical asymptote.
- Understanding structure allows accurate graphing without plotting many points.

Practice Set

Level 1 — Easy

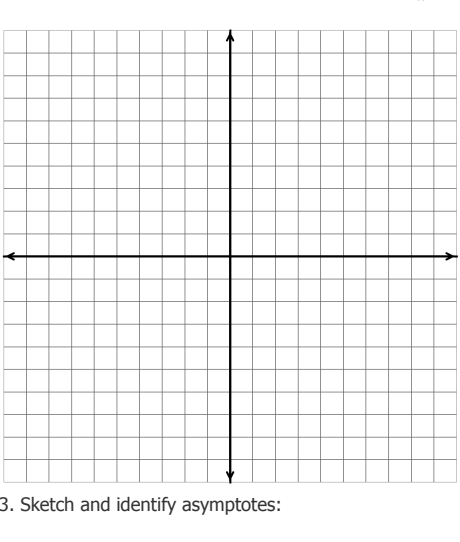
1. Sketch and identify asymptotes:

$$f(x) = \frac{1}{x}$$



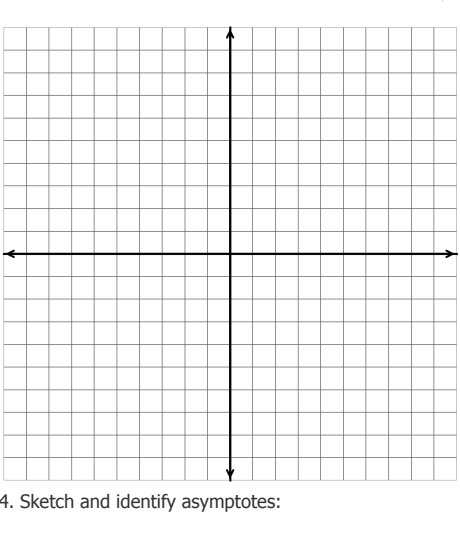
2. Sketch and identify asymptotes:

$$f(x) = \frac{1}{x-1}$$



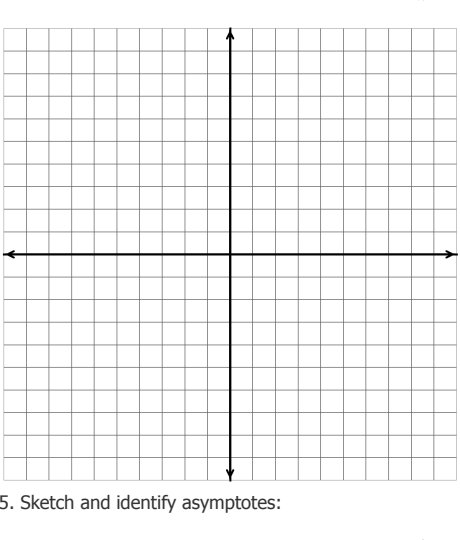
3. Sketch and identify asymptotes:

$$f(x) = \frac{1}{x+3}$$



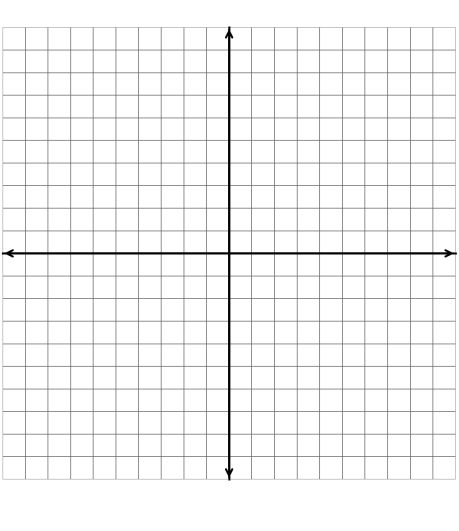
4. Sketch and identify asymptotes:

$$f(x) = \frac{1}{x} - 2$$



5. Sketch and identify asymptotes:

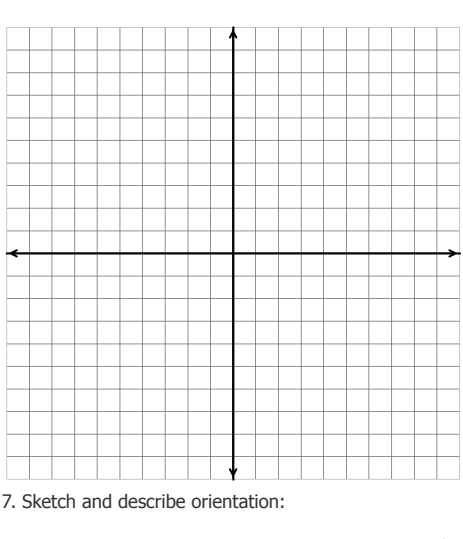
$$f(x) = \frac{1}{x} + 4$$



Level 2 — Medium

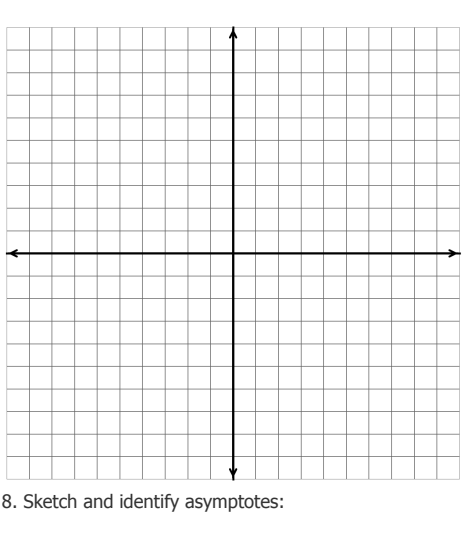
6. Sketch and describe orientation:

$$f(x) = -\frac{1}{x}$$



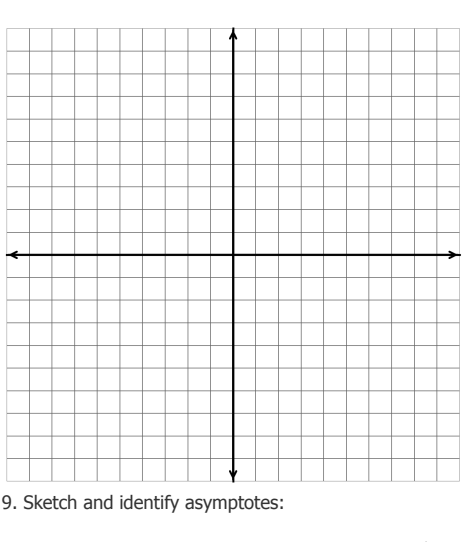
7. Sketch and describe orientation:

$$f(x) = \frac{2}{x}$$



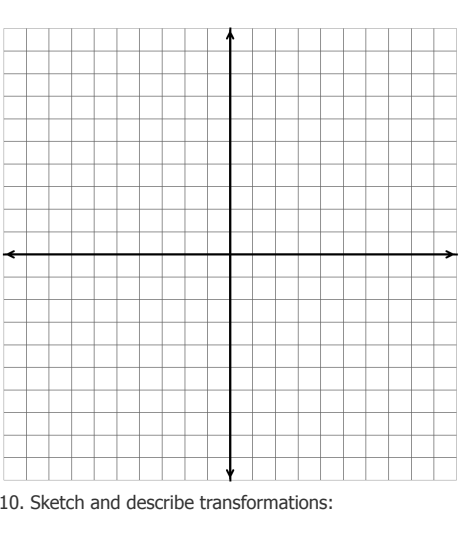
8. Sketch and identify asymptotes:

$$f(x) = \frac{1}{x-2} + 1$$



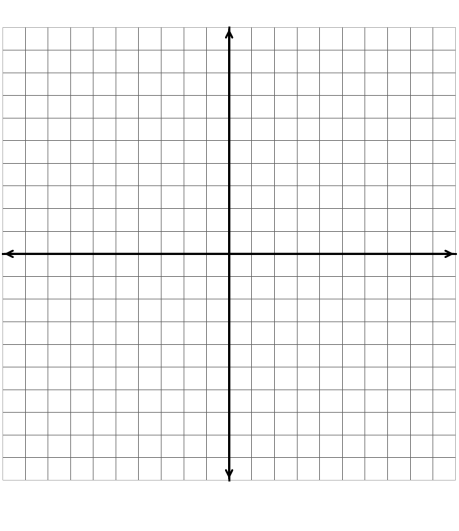
9. Sketch and identify asymptotes:

$$f(x) = \frac{1}{x+1} - 3$$



10. Sketch and describe transformations:

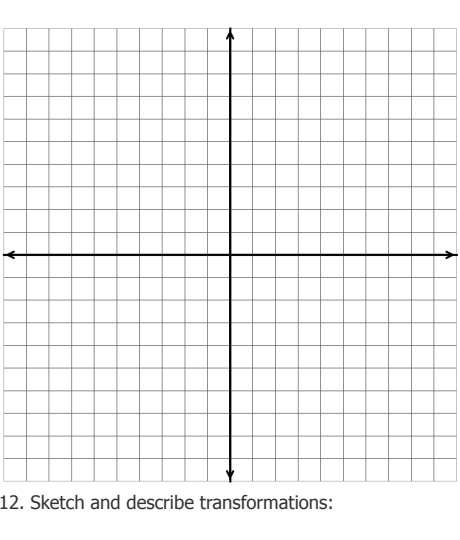
$$f(x) = \frac{3}{x} + 2$$



Level 3 — Hard

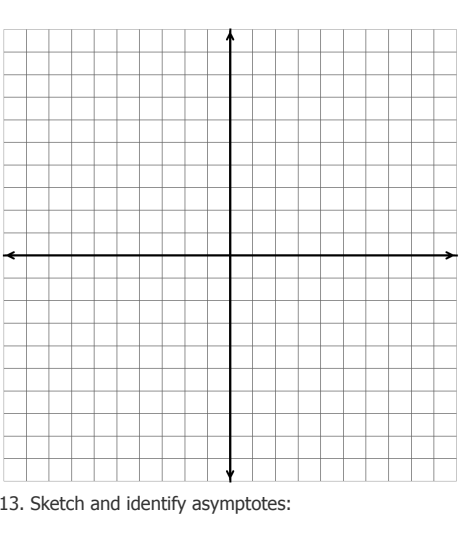
11. Sketch and describe transformations:

$$f(x) = -\frac{2}{x-1}$$



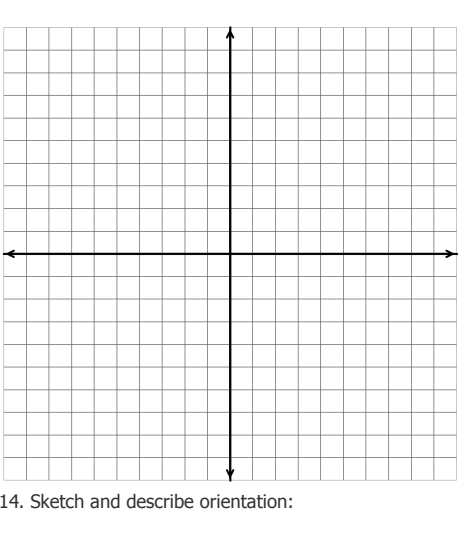
12. Sketch and describe transformations:

$$f(x) = \frac{1}{x-3} - 2$$



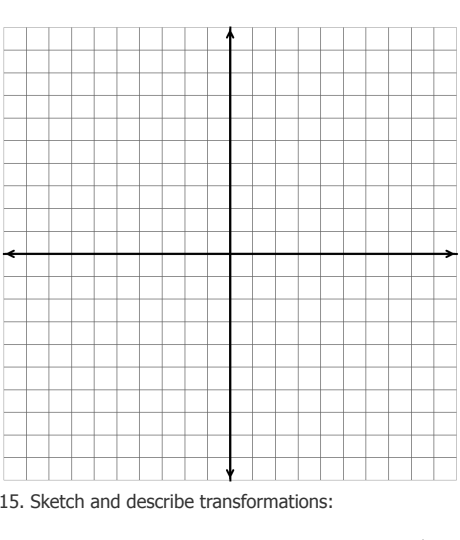
13. Sketch and identify asymptotes:

$$f(x) = \frac{4}{x+2} + 1$$



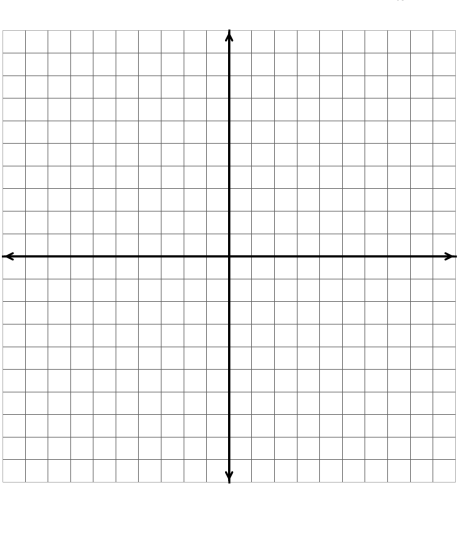
14. Sketch and describe orientation:

$$f(x) = -\frac{5}{x}$$



15. Sketch and describe transformations:

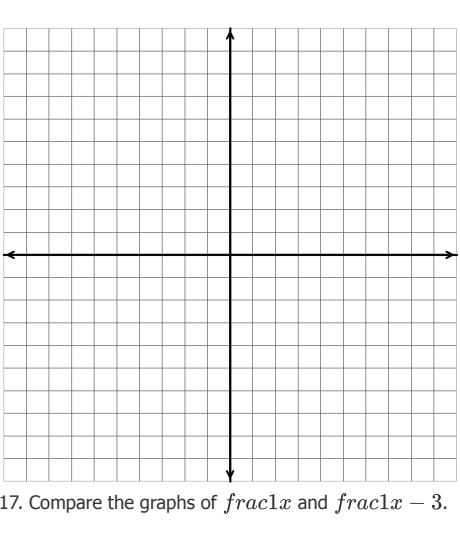
$$f(x) = \frac{1}{x-4} + 3$$



Level 4 — Challenging

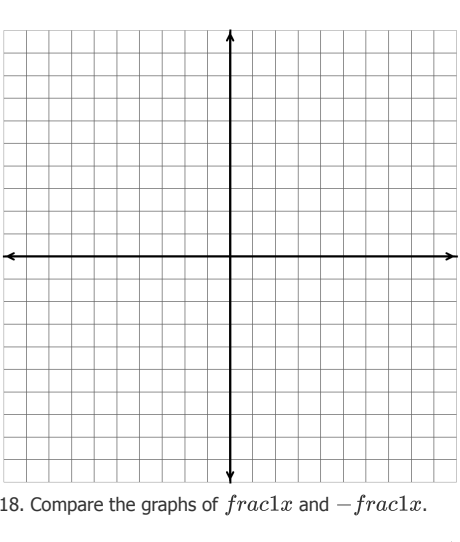
16. Sketch and describe all transformations:

$$f(x) = \frac{-2}{x-2} + 4$$



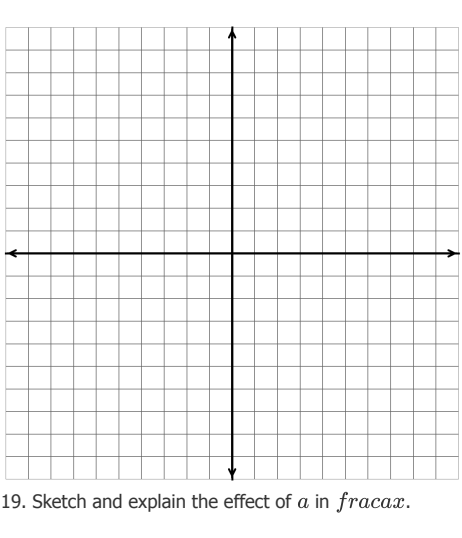
17. Compare the graphs of $frac1x$ and $frac1x - 3$.

$$f(x) = \frac{1}{x-3}$$



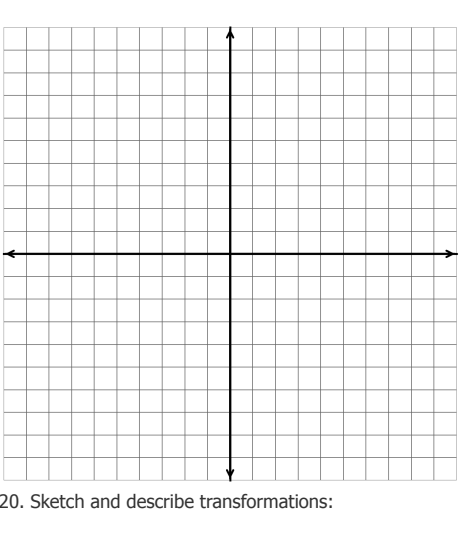
18. Compare the graphs of $frac1x$ and $-frac1x$.

$$f(x) = -\frac{1}{x}$$



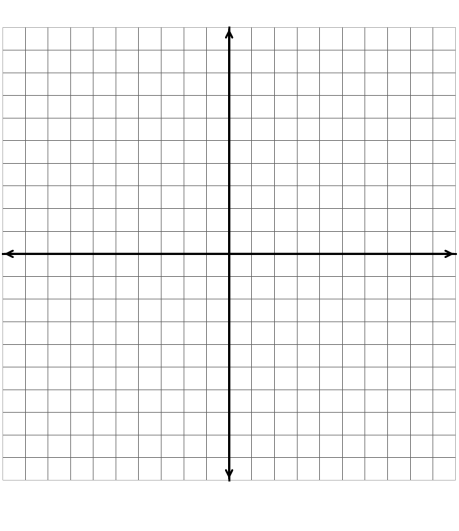
19. Sketch and explain the effect of a in $fracax$.

$$f(x) = \frac{a}{x}$$



20. Sketch and describe transformations:

$$f(x) = \frac{1}{2(x-1)} + 2$$



Level 5 — Justification / Conceptual

21. Explain why reciprocal functions have two branches.

22. Explain why reciprocal graphs never cross vertical asymptotes.

23. Justify how the sign of a affects orientation.

24. Explain why reciprocal functions usually have no intercepts.

25. Describe how asymptotes determine the shape of the graph.

Answer Key

Level 1 — Easy

1. VA: $x = 0$, HA: $y = 0$
2. VA: $x = 1$, HA: $y = 0$
3. VA: $x = -3$, HA: $y = 0$
4. VA: $x = 0$, HA: $y = -2$
5. VA: $x = 0$, HA: $y = 4$

Level 2 — Medium

6. Reflection across x-axis
7. Vertical stretch (factor 2)
8. VA: $x = 2$, HA: $y = 1$
9. VA: $x = -1$, HA: $y = -3$
10. Stretch (factor 3), shift up 2

Level 3 — Hard

11. VA: $x = 1$, reflection, stretch (2)
12. VA: $x = 3$, HA: $y = -2$
13. VA: $x = -2$, HA: $y = 1$, stretch (4)
14. Reflection, stretch (5)
15. VA: $x = 4$, HA: $y = 3$

Level 4 — Challenging

16. VA: $x = 2$, HA: $y = 4$, reflection, stretch (2)
17. Shift right 3 units
18. Reflection across x-axis
19. Controls stretch/compression and orientation
20. VA: $x = 1$, HA: $y = 2$, compression

Level 5 — Justification

21–25. Answers vary (conceptual)

Detailed Example Solutions

Example 1 — Parent Function

$$f(x) = \frac{1}{x}$$

The vertical asymptote is $x = 0$ and the horizontal asymptote is $y = 0$. The graph has two branches in Quadrants I and III.

Example 2 — Horizontal Shift

$$f(x) = \frac{1}{x-2}$$

The graph shifts right 2 units, so the vertical asymptote is $x = 2$ and the horizontal asymptote remains $y = 0$.

Example 3 — Vertical Shift

$$f(x) = \frac{1}{x} + 3$$

The graph shifts upward 3 units. The vertical asymptote remains $x = 0$ and the horizontal asymptote becomes $y = 3$.

Example 4 — Reflection

$$f(x) = -\frac{1}{x}$$

The negative sign reflects the graph across the x-axis, moving branches to Quadrants II and IV.

Example 5 — Vertical Stretch

$$f(x) = \frac{2}{x}$$

The graph is stretched vertically, making it steeper, but asymptotes remain unchanged.

Example 6 — Combined Transformations

$$f(x) = \frac{1}{x+4} - 1$$

The graph shifts left 4 units and down 1 unit. The asymptotes become $x = -4$ and $y = -1$.

Example 7 — Reflection and Translation

$$f(x) = -\frac{3}{x-1} + 2$$

The graph shifts right 1 unit and up 2 units. The negative sign reflects the graph and the factor of 3 creates a vertical stretch.